ShortageGraph: Recall Exposure and Market Structure in the U.S. Drug Shortage Snapshot, and the Limits of Public Onset Labels

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Abstract—Hospital pharmacists usually learn of a drug shortage only after it is posted, so predicting which drugs will be listed next is an attractive target for clinical informatics. Published models are tabular and supervised on historical posting labels, which presumes that history is complete. We test that presumption and then ask what the public record can support instead. First, we build the U.S. drug supply network from the National Drug Code directory: 10,537 drug nodes with 88,071 edges at the median week, of which 68.0% have a single labeler. Second, we conduct a cross-sectional study at a single reference date (22 October 2026), comparing 293 drugs on the FDA shortage list with 14,313 marketed drugs not on it. A recall of the same product in the preceding 24 months was present for 17.1% of listed drugs versus 2.5% of unlisted ones (odds ratio 8.07, 95% CI 5.85–11.13); because cases are ascertained through product codes, we re-estimate within product-count strata, where the pooled odds ratio falls to 3.64 and four of five strata still exclude unity. The association holds for injectables (14.1% versus 3.7%, odds ratio 4.29). Finally, we show that the openFDA shortage mirror holds 1,142 current entries but only 19 resolved ones, so first-posting labels are survivorship-biased; a released gate refuses to train on uncertified years. Finally, we specify the forecasting system those labels were meant to train —relational message passing, a monotone hazard head, occlusion attribution and a 25-drug weekly review list—and report no performance for it, because a number computed on those labels would describe the bias. Code for all of it is released.

Index Terms—drug shortages, supply network, cross-sectional study, graph neural networks, reproducibility

I. Introduction

Drug shortages are a persistent threat to patient care, with well documented clinical cost: delayed and cancelled therapy, medication errors and forced switches to less familiar agents [2]. The American Society of Health-System Pharmacists counted 227 active national shortages at the end of the second quarter of 2026, down from 323 in early 2024 but rising for consecutive quarters, and just under half of new shortages involved single-manufacturer products [1]. The FDA attributes most shortages to manufacturing and quality problems concentrated in few facilities [3], which makes the supply network, not any single drug, the natural unit of analysis.

Prediction work in this area is young. Pall et al. [4] built supervised models from the sales records of 22 Canadian pharmacies; Roe et al. [5] modelled duration and cause from 1,054 Korean reports. Both treat each drug as an independent row. Eberhardt et al. [6] constructed a descriptive knowledge graph of the German supply chain to find critical suppliers, and Kosasih et al. [7] applied graph neural networks to supply chain risk, but neither forecasts shortages. All supervised approaches share one dependency: a trustworthy historical label of when each drug entered shortage.

That dependency would be worth the trouble because shortage risk is relational: a quality event at one facility threatens every product made there, and a drug with few labelers is exposed through each of them. Those two-hop paths carry signal that a row of per-drug counts cannot represent.

This paper started as an attempt to build that label for the United States and ended by testing it. Our contributions are:

1) A measured network. A weekly-dated graph of U.S. drug products and the labelers that market them, reconstructed from the National Drug Code directory [8] over 96 sampled weeks, with the concentration statistics that motivate relational analysis (Table I).
2) A cross-sectional association study. At one reference date, drugs on the FDA shortage list are compared with all other marketed drugs on two exposures that need no shortage history: whether the same product was recalled recently, and how many labelers market it (Tables II and III).
3) A label-provenance audit. We quantify why the openFDA shortage record cannot support first-onset forecasting, and release a gate that refuses to train on years it cannot certify, so the failure is caught rather than silently learned.

TABLE I
Measured network structure. Values are medians over 96 monthly snapshots, 2018–2025, computed from 114,018 NDC products.

Quantity	Value
Sampled weekly snapshots	96
Drug nodes $\lvert V_D \rvert$	10,537
Labeler nodes $\lvert V_L \rvert$	4,391
Edges (markets + substitutes)	88,071
Drugs with a single labeler	68.0%
Median labelers, drugs on shortage list	4
Median labelers, drugs not on list	1

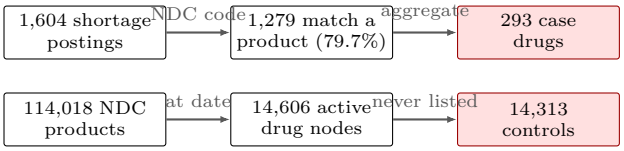


Fig. 1. How cases and controls are assembled at the reference date. A case enters only when a posting carries an NDC that resolves to a marketed product, so a drug with many listed products is easier to capture and easier to mark exposed. That advantage is what the product-count stratification in Table III controls.

4) A released system, and no claim about it. The relational model, monotone hazard head, attribution rule and evaluation harness are specified, implemented and unit-tested (Section V), with no test performance reported, for the reason Section IV establishes.

Clinical relevance—A recall of the same product is the kind of signal a pharmacy already receives, weeks or months before a shortage appears on a national list. Knowing that this signal is enriched among listed drugs, and by how much, tells a procurement pharmacist where to look first, and tells the informatics community which labels are safe to learn from.

II. Data and Study Design

A. Supply Network from the NDC Directory

Every marketed U.S. drug product carries a National Drug Code. We parse the directory [8] into 114,018 products and aggregate them into drug nodes keyed by active ingredient set, dosage form and route, which matches the unit in which FDA shortage entries are reported. A product is active at a date if its marketing start has passed and its marketing end has not. Two relations are built per snapshot: markets (drug–labeler) and substitutes (drug–drug, same ingredients with a different form or route).

Manufacturing establishments are deliberately absent: linking products to FDA Establishment Identifiers requires the NDC-to-FEI mapping from DailyMed Structured Product Labeling [10], which this study does not claim to have completed. Counts reported here describe product and labeler structure only, and no result uses two-hop facility paths.

B. Reference Date, Cases and Controls

The shortage records come from the openFDA shortage endpoint [9], which mirrors the FDA Drug Shortages Database [11]. We take the latest initial posting date present, 22 October 2026, as the single reference date; using one date for every subject is what makes the design interpretable, because exposure windows then have a common end.

A drug is a case if any shortage record maps into the network at the reference date (current or to-be-discontinued status), and a control if it is active in the directory then and never appears in the shortage records. Of 1,604 records, 1,279 (79.7%) carry an NDC that resolves to a study drug, yielding 293 distinct case drugs; 325 (20.3%) could not be mapped and are excluded, and controls number 14,313. This is a prevalence case-control study, reported as a cross-sectional association; it estimates neither incidence nor causal effect. Fig. 1 shows how the groups are assembled, and the asymmetry in how they are found is why Table III exists.

C. Exposures

Two exposure definitions are used, and they are kept separate because they differ in specificity.

Same-product recall. An enforcement report [9] published in the 24 months before the reference date lists an NDC belonging to the drug node: precise, because it identifies a recall of that product.

Any-labeler recall. An enforcement report in the same window names a firm that markets the drug — broad, and, as Section VI notes, partly a measure of firm size.

Concentration. The number of distinct labelers behind a drug, from the directory alone.

Of 1,587 enforcement reports in the window, 808 carry an NDC that resolves to a study drug. Because cases and exposures are both reached through NDC codes, a drug with many products is easier to capture and easier to mark exposed, so we treat product count as an ascertainment variable and stratify on it (Section II-D); this is the study's most important correction.

D. Statistics

Associations are reported as odds ratios with Katz 95% confidence intervals and two-sided Fisher exact tests, computed from the hypergeometric distribution without approximation [12]. Results are pooled across strata with the Mantel–Haenszel estimator. Strata are defined two ways: by concentration (single versus multiple labelers) and by product count (1, 2–3, 4–7, 8–20, 21 or more NDCs). The second is the ascertainment control; the first tests whether concentration explains the recall association. Sensitivity analyses repeat the exposure with 12- and 36-month windows. All computations are in shortagegraph.casecontrol, which reports the 2×2 table behind every estimate.

TABLE II
Cross-sectional associations at 22 October 2026 (293 cases, 14,313 controls; 24-month exposure window). OR: odds ratio; CI: Katz 95% confidence interval; p: two-sided Fisher exact.

Exposure	Cases	Controls	OR	95% CI	p
Same-product recall	17.1%	2.5%	8.07	5.85–11.13	1.7×10^{-25}
Any-labeler recall	84.6%	22.9%	18.55	13.47–25.55	2.0×10^{-107}
Either definition	85.3%	23.3%	19.18	13.85–26.57	2.2×10^{-108}
Single labeler	22.5%	67.8%	0.14	0.11–0.18	1.1×10^{-54}

TABLE III
Same-product recall, stratified by number of NDC products per drug (ascertainment control). Pooled Mantel–Haenszel OR 3.64.

NDC products	Cases	Controls	OR	95% CI	p
1	34	7,581	29.60	9.89–88.59	2.2×10^{-5}
2–3	60	2,671	8.07	3.47–18.77	8.3×10^{-5}
4–7	48	1,470	2.38	0.83–6.87	0.107
8–20	87	1,618	4.17	2.29–7.61	3.3×10^{-5}
21+	64	973	2.53	1.45–4.42	0.002

III. Results

Table II reports the crude associations. A recall of the same product in the preceding 24 months is present for 17.1% of listed drugs against 2.5% of unlisted ones (OR 8.07, 95% CI 5.85–11.13), and the broad firm-level definition for 84.6% against 22.9% (OR 18.55). At 12 and 36 months the same-product odds ratio is 8.97 and 8.31, so the finding does not depend on the window.

A. Ascertainment and the Concentration Result

The crude concentration estimate points the other way from the premise that motivates most shortage work: listed drugs are less often single-labeler (22.5% versus 67.8%, OR 0.14), with a median of four labelers against one. We do not read this as evidence that competition protects supply: cases are found by matching product codes, so a drug with many products is more likely to be captured, and the same mechanism inflates any NDC-based exposure. It is therefore reported as an ascertainment artefact, not a clinical finding.

The correction matters quantitatively. Within product-count strata the pooled Mantel–Haenszel odds ratio falls from 8.07 to 3.64, so roughly half of the crude association was the capture advantage of large products. Four of five strata still exclude unity, including the 21-or-more stratum (OR 2.53, 95% CI 1.45–4.42): attenuated, not removed.

B. Concentration Does Not Explain the Signal

Stratifying the recall association by concentration gives odds ratios of 14.06 (95% CI 8.61–22.96) among single-labeler drugs and 13.11 (7.47–23.01) among multi-labeler drugs, so the signal survives that stratification; only the product-count stratification attenuates it, and only partially.

C. Injectable Subgroup

Injectables are the class whose absence most disrupts acute and oncology care. Among 156 injectable case and 2,224 control drugs a same-product recall is present for 14.1% against 3.7% (OR 4.29, 95% CI 2.60–7.08), and the broad definition for 87.2% against 32.9%. so the association holds where a procurement response is feasible.

IV. What the Public Record Cannot Support

The study above was designed after the forecasting design failed, and the reason belongs in the record. First-posting labels require a complete history: a drug missing from it looks never-listed, which turns an existing shortage into a false negative rather than merely removing it.

The openFDA shortage mirror is a live snapshot, not an archive. Of 1,604 records, 1,142 are current, 443 are to be discontinued and only 19 are resolved. A shortage that began in 2019 and resolved in 2021 is simply absent. Collapsing to distinct drugs leaves 307, and per-year first-posting counts over the intended training window are 13 (2018), 4 (2019), 17 (2020), 11 (2021) and 18 (2022), against the hundreds the FDA recorded annually.

We did not attempt to repair this by modelling. Instead, shortagegraph.labels computes coverage per year, checks that postings match the network's keys, and refuses to certify a window below a stated floor; train.py exits rather than training on an uncertified window. Both splits fail certification, and the released code reproduces that failure. The obvious escape routes are closed: the Wayback Machine has no usable snapshots of the legacy shortage listing page, and the FDA Data Dashboard returns HTTP 403 to automated clients. A complete history would have to come from an FDA archival export, a commercial shortage dataset, or a partner health system's own records.

V. The System These Labels Would Train

The audit above says which labels are unavailable; this section specifies what would be fitted to them. Most of the specification does not depend on the labels, and every component named here exists in the released code and is covered by the test suite.

A. Graph and Dated Features

A snapshot at week t is a heterogeneous graph $G_t = (V_D \cup V_L, \{E_r\}_{r\in R}, X_t)$ with drug nodes V_D and labeler nodes V_L. Two relations are built: markets (drug–labeler) and substitutes (drug–drug, same ingredients, different form or route), each with reverse edges so information flows both ways. Facility nodes are absent, because the NDC-to-FEI mapping is not built, so with $K = 2$ layers a drug reaches its co-marketed products through a shared labeler. That is a weaker version of the two-hop facility path the design was built around, and no result here depends on it. All features come from records published on or before t, over trailing windows of 13, 26 and 52 weeks, keyed to disclosure dates rather than event dates so that no feature precedes publication; recalls contribute counts by class and weeks since the most recent report.

B. Monotone Hazard Head and Attribution

For $h \in \{4, 8, 12\}$ weeks the label is $y_{v,t}^{(h)} = \mathbf{1}[t < \tau_v \leq t+h]$ with τ_v the week of first posting, restricted to drugs never yet listed, and snapshots whose label windows would cross a split boundary are embargoed — a rule config.py implements and the suite checks for leakage. Fitting three independent classifiers can return a higher 4-week than 12-week risk, so we model a discrete hazard over $(0, 4]$, $(4, 8]$ and $(8, 12]$ instead,

$$\lambda_{v,j} = \sigma(w_j^\top h_v + b_j), \qquad p_v^{(h_m)} = 1 - \prod_{j=1}^{m} \big(1 - \lambda_{v,j}\big), \quad (1)$$

which is monotone in m by construction. The encoder is relational message passing with mean aggregation [13], [14],

$$h_v^{(k+1)} = \sigma\Big(W_0^{(k)} h_v^{(k)} + \sum_{r \in R} \sum_{u \in N_r(v)} \frac{1}{|N_r(v)|} W_r^{(k)} h_u^{(k)}\Big), \quad (2)$$

trained on class-weighted cross-entropy with $\alpha = \sqrt{n_-/n_+}$, the square root because the full ratio overweighted the few positives in development. Baselines see the same drug-level features, with labeler signals preaggregated as counts and maxima: a sole-source rule, L_2 logistic regression, gradient-boosted trees [15] and an edge-free MLP, the last isolating the value of structure from model capacity. An alert is explained by occlusion, $\Delta_{u \to v} = p(G_t) - p(G_t \setminus u)$, which replaces one upstream node's features by their training median while leaving topology intact; occlusion is preferred to a learned mask such as GNNExplainer [17] because a pharmacist can read it directly.

C. Protocol, and Why Nothing Is Reported

Positives are rare, so the primary metric is AUPRC [16], with AUROC, precision in a weekly 25-drug review list sized to pharmacy workload, the Brier score, bootstrap intervals resampling drugs rather than drug-weeks, lead time before posting, and an injectable subgroup. The harness implements each, and the pipeline runs end to end on a synthetic fixture that verifies mechanics only: shapes, label semantics, absence of future leakage, metric agreement with reference implementations, and monotonicity of the hazard head. We report no test performance, and the omission is the point. Section IV shows the label history is survivorship-biased, so a model fitted to it would learn which shortages were recorded, not which drugs failed; train.py refuses to fit on an uncertified history rather than emitting such a number.

VI. Discussion and Limitations

The central limitation is design: a single cross-section estimates who is in shortage at one date, not who will be. It cannot show that a recall precedes an onset, and for a drug listed for years a 2025 recall may follow the shortage rather than lead it. Reverse causation is therefore compatible with our numbers, and the finding is best read as co-occurrence, consistent with the FDA's root-cause analysis [3] but not proof of it.

The broad exposure definition is nearly a measure of firm size, since a large labeler has many recalls and many products; we report it for completeness and rely on the same-product definition, which is narrow enough to mean something.

Case ascertainment is the other constraint: because cases are reached through product codes, the concentration estimate and part of the recall estimate are shaped by how many products a drug has, which the stratified analysis quantifies rather than removes. A study with an independent case source, such as a health system's shortage log, would not share the bias.

Public data also omit what determines supply — ingredient sourcing, inventory, contracts and demand surges — so a public-data association is a floor, not a ceiling. And the shortage list reflects reporting behaviour as much as scarcity, so a study keyed to one list partly models that list.

The practical implication is nonetheless usable. A recall of the same product is already known to a pharmacy when it happens, and it is enriched among drugs that appear on the national shortage list. That supports using recall history to prioritise which products a procurement pharmacist checks first, within existing allocation rules, while the forecasting question waits for labels that can be certified.

Two limits are specific to the system in Section V. Deployment carries a risk the design creates: if many institutions stockpile on one alert, the forecast can cause the shortage it predicts, so a review list belongs inside existing allocation rules and beside stewardship, not in a purchasing trigger. And a retrospective design cannot show that earlier warning improves outcomes; the step that could is a silent prospective evaluation, with alerts logged but not acted on and scored against a partner's own shortage records.

VII. Conclusion

We rebuilt the U.S. drug supply network from public listing data and asked what it can support. Cross-sectionally, same-product recall exposure is strongly associated with being on the shortage list (OR 8.07 crude, 3.64 after stratifying on product count, 4.29 in injectables), while the sole-source premise is not recoverable from list-based labels, because case ascertainment favours large products. First-onset labels cannot be certified, so we release the gate that says so rather than a model trained on corrupt negatives, plus the relational model and protocol a certifiable source would train: the design survives its own audit as a reproducible benchmark, not a withdrawn claim. Code is at https://github.com/ohtherewegoagain/shortagegraph.

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References

[1] American Society of Health-System Pharmacists and University of Utah Drug Information Service, "National drug shortages: January 2001 to June 2026," ASHP Drug Shortages Statistics, 2026. [Online]. Available: https://www.ashp.org/drug-shortages/shortage-resources/drug-shortages-statistics

[2] M. McLaughlin, D. Kotis, K. Thomson et al., "Effects on patient care caused by drug shortages: a survey," J. Manag. Care Pharm., vol. 19, no. 9, pp. 783–788, 2013.

[3] U.S. Food and Drug Administration, Drug Shortages Task Force, "Drug shortages: Root causes and potential solutions," FDA, Silver Spring, MD, USA, 2019.

[4] R. Pall, Y. Gauthier, S. Auer, and W. Mowaswes, "Predicting drug shortages using pharmacy data and machine learning," Health Care Manag. Sci., vol. 26, no. 3, pp. 395–411, 2023.

[5] H. S. Roe et al., "Drug shortage in South Korea: machine learning-based prediction models and analysis of duration and causal factors," Front. Pharmacol., vol. 16, art. 1608843, 2025.

[6] K. Eberhardt, A. Schwärzel, F. K. Kaiser, S. Rosenberg, and F. Schultmann, "Leveraging knowledge graphs in pharmaceutical supply chains: insights into key drivers of drug shortages," Int. J. Prod. Res., vol. 63, no. 19, pp. 7129–7152, 2025.

[7] E. E. Kosasih, F. Margaroli, S. Gelli, A. Aziz, N. Wildgoose, and A. Brintrup, "Towards knowledge graph reasoning for supply chain risk management using graph neural networks," Int. J. Prod. Res., vol. 62, no. 15, pp. 5596–5612, 2024.

[8] U.S. Food and Drug Administration, "National Drug Code Directory." [Online]. Available: https://www.fda.gov/drugs/drug-approvals-and-databases/national-drug-code-directory

[9] T. A. Kass-Hout et al., "OpenFDA: an innovative platform providing access to a wealth of FDA's publicly available data," J. Am. Med. Inform. Assoc., vol. 23, no. 3, pp. 596–600, 2016.

[10] U.S. National Library of Medicine, "DailyMed." [Online]. Available: https://dailymed.nlm.nih.gov

[11] U.S. Food and Drug Administration, "FDA Drug Shortages Database." [Online]. Available: https://dps.fda.gov/drugshortages

[12] K. J. Rothman, S. Greenland, and T. L. Lash, Modern Epidemiology, 3rd ed. Philadelphia, PA, USA: Lippincott Williams & Wilkins, 2008.

[13] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in Proc. ESWC, 2018, pp. 593–607.

[14] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in Proc. NeurIPS, 2017, pp. 1024–1034.

[15] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in Proc. ACM SIGKDD, 2016, pp. 785–794.

[16] T. Saito and M. Rehmsmeier, "The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLoS ONE, vol. 10, no. 3, e0118432, 2015.

[17] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: generating explanations for graph neural networks," in Proc. NeurIPS, 2019, pp. 9240–9251.